

Geometrical factors and interfacial processes affecting complex dielectric permittivity of partially saturated porous media

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[1] Methods for remote and ground-based sensing of soil water content often rely on determination of bulk dielectric permittivity which may be affected by factors other than water content, such as interfacial polarization (Maxwell-Wagner effect), phase configuration, and electrical conductivity. The complex interactions among these factors were modeled by the Maxwell-Wagner-Bruggeman-Hanai formalism on the basis of self-consistent differential effective medium approximation. The modeling framework considers porous media composed of coated inclusions and enables systematic evaluation of various factors on determination of water content at different measurement frequencies associated with different sensors. Simulation results illustrate the importance of accounting for phase configuration; using a simple saturation-dependent weight function and only two typical phase configurations provided good agreement with a wide range of dielectric measurements spanning different porous media, saturations, and measurement frequencies. The model provides the necessary framework for incorporation of the Maxwell-Wagner effect on standard measurements obtained by time domain reflectometry and through frequency response enables evaluation and a wide array of other sensors. Work is under way to incorporate effects of ambient conditions (temperature) and of bound water on bulk dielectric measurements.

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1. Introduction

[2] Reliable and timely measurement of soil's water content is important for many hydrologic and climatic models and for agricultural water management. Many nondestructive methods of water content measurement rely on electromagnetic determination of bulk dielectric permittivity of the wet soil, capitalizing on the relatively high dielectric permittivity (~ 80) of water relative to that of other constituents (e.g., ~ 5 for soil minerals, and 1 for air). This is evidenced by the success of time domain reflectometry (TDR) [Topp *et al.*, 1980; Dalton *et al.*, 1984] in a host of hydrological and agricultural applications, and rapid expansion of related sensing techniques. A key factor in the success of TDR is attributed to early claims of universal relationships between measured bulk dielectric permittivity and water content, based on empirical relationships proposed by Topp *et al.* [1980].

$$\theta = 4.3 \times 10^{-6} \epsilon_b^3 - 5.5 \times 10^{-4} \epsilon_b^2 + 2.92 \times 10^{-2} \epsilon_b - 5.3 \times 10^{-2} \quad (1)$$

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[3] Early applications of Topp's equation confirmed its adequacy for many mineral soils; however, as more measurements became available, limitations of the empirical relationships for soils with high organic matter and high clay content became apparent [Dirksen and Dasberg, 1993]. Theoretical and experimental evidence suggests that factors other than water content, including soil texture, salinity, ambient temperature and frequency (not considered in Topp's empirical equation), affect bulk permittivity. Specifically, studies have documented influences of interfacial processes [Alvarez, 1973; de Lima and Sharma, 1992; Chelidze and Gueguen, 1999; Hilhorst, 1998], geometrical microstructures [Sen, 1981; Knight and Endres, 1990; Garrouch and Sharma, 1994; Gueguen *et al.*, 1997; Endres, 2000; Robinson and Friedman, 2001; Jones and Or, 2002] and ambient conditions such as temperature [Or and Wraith, 1999], all of which are known to vary with measurement frequency. Advances in sensing technology and the introduction of new sensors operating at different frequencies require a more general modeling framework capable of distinguishing among the various factors and quantify their relative contributions to measured bulk dielectric permittivity with explicit consideration of a sensor's measurement frequency. In soils, often characterized by large interfacial area per unit volume, the role of interfacial polarization is of particular interest because of its dominance of bulk dielectric response at low measurement frequency (MHz) where many new sensors operate.

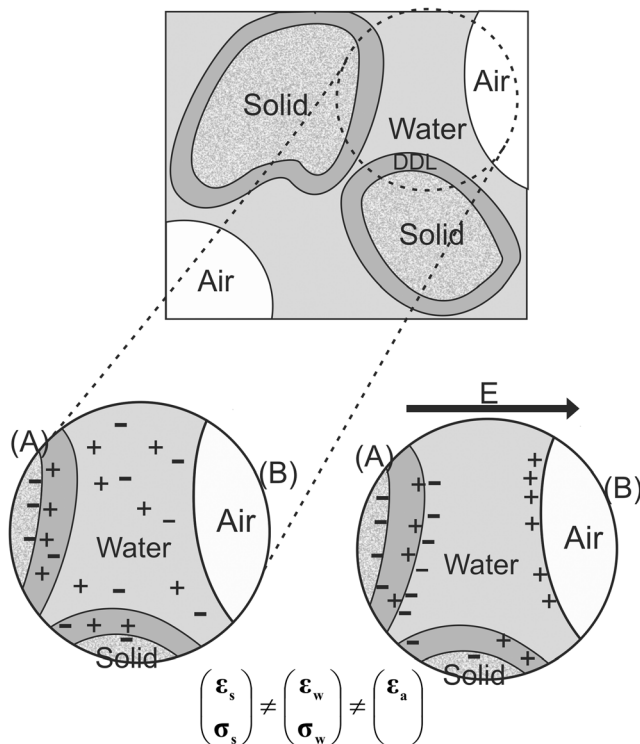


Figure 1. Interfacial polarization (M-W effect) in soil due to perturbation of ionic charges: alignment of charges associated with solid-liquid-air interface by applied electrical field (A) and accumulation of charges at the solid-liquid-air interface due to mismatch in electrical conductivity between the phases (B).

1.1. Maxwell-Wagner Effect

[4] The extent and complexity of spatial arrangement of nonconducting soil solids and air fragment the electrically conductive soil solution, which gives rise to bulk interfacial polarization [Chelidze and Gueguen, 1999], also termed as the Maxwell-Wagner (M-W) effect. The M-W effect occurs in heterogeneous mixtures where discontinuities (interfaces) separate materials with different dielectric permittivity and electrical conductivity (as shown in Figure 1), yet the normal component of the electrical field must be equal across the interface [Alvarez, 1973; Chelidze and Gueguen, 1999]. Consequently, differences in electrical properties result in mismatch in current densities and interfacial charging (charge accumulation at interfaces) leading to enhancement of bulk dielectric permittivity (higher than linear combination of its components). Typically these processes remain active in the frequency range of <100 MHz. Interfacial charging occurs at an exponential rate with a characteristic time constant or frequency that marks the relaxation of these interfacial processes. The relaxation frequency for the M-W effect depends on the bulk electrical conductivity of the saturating solution.

1.2. Physical Model and Phase Configurations

[5] A considerable body of theoretical work has focused on the estimation of bulk permittivity of heterogeneous media with consideration of the microstructure and M-W effect [Sen et al., 1981; Norris, 1985; Hashund et al., 1994;

Friedman, 1997; Gueguen et al., 1997; Endres, 2000; Asami, 2002; Cosenza et al., 2003]. Differential effective medium approximation (DEMA) [Bruggeman, 1935; Sen et al., 1981; Norris, 1985; Endres, 2000; Cosenza et al., 2003] represents a popular and well-established modeling approach that seems to provide best agreement with experimental data [Cosenza et al., 2003]. For a two-phase mixture, the DEMA designates one phase as a background matrix to which the other phase is added incrementally as inclusions [Hanai, 1968; Norris et al., 1985; Chelidze and Gueguen, 1999]. For a three-phase mixture, resembling more closely the three phases in unsaturated porous media (solid, water, air), two different extensions based on two-phase system are available. One extension adds two phases incrementally as separate inclusions to a background of the third phase [Norris, 1985]. The sequence by which components are added plays an important role in the resulting homogenized background permittivity for the next step [Cosenza et al., 2003]. For example, adding soil solids then air into a water background, or adding air first then solids, yields different values of effective permittivity. Another extension to the DEMA method considers incremental addition of coated inclusions (one phase coating the other) to a background of the third phase [Asami, 2002]. The inclusion-based DEMA formulation and associated assumptions for the application used in this study was termed by Chelidze and Gueguen [1999] as the Maxwell-Wagner-Bruggeman-Hanai (MWBH) theory. The MWBH formalism will be implemented on the basis of Asami's [2002] notation considering natural associations and configurations among soil phases (e.g., water-solid association). Sen et al. [1981] pointed out the advantage of considering common association between rock hydrophilic surfaces with the aqueous phase which significantly reduces the number of hypothetical phase configurations to be considered for modeling [Friedman, 1997]. Despite shortcomings of the MWBH formulation mainly those associated with "bulk" phase representation disregarding surface conductivity and inclusion clustering, Chelidze and Gueguen [1999] asserted that "The MWBH theory is a starting point for any theoretical interpretation of experimental electrical spectra."

[6] The primary objectives of the study were (1) to incorporate M-W effect and phase configuration into dielectric models used for interpretation of hydrological and geophysical measurements of water content and electrical conductivity using the MWBH DEMA formalism; (2) to test the frequency-dependent dielectric permittivity predictions by the model using experimental data from a wide range of porous media reported in the literature; and (3) to deduce from model performance general trends regarding model suitability for various low- and high-frequency dielectric measurement interpretation.

2. Theoretical Considerations

2.1. MWBH-Asami's Differential Effective Medium Approximation Model

[7] Differential effective medium approximation (DEMA) is an inclusion-based method that considers the bulk properties and phase configuration of the components of the porous medium. Assuming that a porous medium is constituted by two spherical components m and p with corresponding

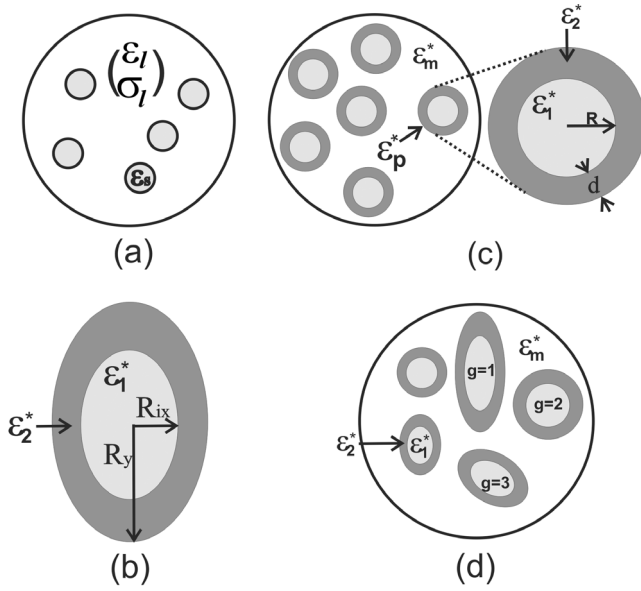


Figure 2. (a) Wagner's two-phase model with low concentration inclusion, (b) definition sketch of single ellipsoidal inclusion coated by a shell used for Hanai-Asami's DEMA three-phase system, (c) Hanai-Asami's DEMA adding coated inclusions into the background matrix at each step, and (d) different inclusion group g divided by different inclusion shape.

permittivity ϵ_m^* and ϵ_p^* , volume u and v , and the volumetric fractions $1 - \phi$ and ϕ to calculate the effective permittivity of this medium, it starts with designating one component as the matrix material for the initial condition, for example, m as the matrix material. Initially an infinitesimal amount of material p (Δv) is added into material m . This is followed by homogenization of the mixture and obtaining an effective permittivity ϵ_i^* . The homogenized mixture is then considered as the “new” matrix and previous step is repeated by addition of Δv inclusions. In this process, the effective permittivity immediately after the i th step is [Hanai, 1968; Sen et al., 1981]

$$\frac{\epsilon_i^* - \epsilon_{i-1}^*}{\epsilon_i^* + 2\epsilon_{i-1}^*} = \frac{\Delta v}{u + v} \frac{\epsilon_p^* - \epsilon_{i-1}^*}{\epsilon_p^* + 2\epsilon_{i-1}^*} \quad (2)$$

[8] Keeping in mind the permittivity ϵ_i^* of the new mixture is not much different from that of the matrix ϵ_{i-1}^* since the added amount of inclusion is infinitesimal ($\Delta v \rightarrow 0$). Taking the limit enables to express equation (2) in differential form as

$$\frac{d\epsilon^*}{3\epsilon^*} = \frac{dv}{u + v} \frac{\epsilon_p^* - \epsilon^*}{\epsilon_p^* + 2\epsilon^*} \quad (3)$$

[9] Substitution of volumes with volumetric fractions, results in

$$\frac{d\epsilon^*}{3\epsilon^*} = \frac{d\Psi}{1 - \Psi} \frac{\epsilon_p^* - \epsilon^*}{\epsilon_p^* + 2\epsilon^*} \quad (4)$$

[10] Solving equation (4) with boundary conditions $\Psi|_0^\phi$ and $\epsilon^*|_{\epsilon_m^*}^{\epsilon_p^*}$ yields

$$\left(\frac{\epsilon^* - \epsilon_p^*}{\epsilon_m^* - \epsilon_p^*} \right) \left(\frac{\epsilon_m^*}{\epsilon^*} \right)^{1/3} = 1 - \phi \quad (5)$$

[11] The above description for a two-phase system is extended to a three-phase system by simply treating added inclusions as confocal inclusions with a core coated with a shell of a third phase [Asami, 2002] (as shown in Figure 2c). At each step, an infinitesimal amount (Δv) of two-phase coated inclusions is added to the background and homogenized as described above. The final mixture's effective permittivity ϵ^* for a three phase mixture is given by

$$\begin{aligned} \left(\frac{\epsilon^* - \epsilon_p^*}{\epsilon_m^* - \epsilon_p^*} \right) \left(\frac{\epsilon_m^*}{\epsilon^*} \right)^{1/3} &= 1 - \phi \\ \epsilon_p^* &= \epsilon_2^* \frac{2(1 - \lambda)\epsilon_2^* + (1 + 2\lambda)\epsilon_1^*}{(2 + \lambda)\epsilon_2^* + (1 - \lambda)\epsilon_1^*} \\ \lambda &= \left(\frac{R}{R + d} \right)^3 \end{aligned} \quad (6)$$

where d is the thickness of the shell, R is the radius of the core, ϵ_1^* and ϵ_2^* are the complex permittivities of the core and shell respectively (Figure 2c).

[12] Accounting for variable inclusion geometry, we consider a distribution of different shapes of ellipsoidal inclusions in a mixture with matrix dielectric permittivity (ϵ_m^*) and inclusions (ϵ_p^*), where the ellipsoidal core (ϵ_1^*) is coated by the ellipsoidal shell (ϵ_2^*) (Figure 2b). We define the semi-axes of ellipsoidal core by R_{ik} ($k = x, y, z$), and that of the ellipsoidal inclusion by R_k ($k = x, y, z$) (Figure 2b). Inclusions may be classified to different shape groups g having different R_{ik} and/or R_k . The mathematical formulation of such geometrical extension is

$$\begin{aligned} \int_0^\phi -\frac{d\phi'}{1 - \phi'} &= \int_{\epsilon_m^*}^{\epsilon^*} \frac{3}{\epsilon^*} \left[\sum_{g=1}^n r_g \sum_{k=x,y,z} \frac{\epsilon^* - \epsilon_{pgk}^*}{(\epsilon_{pgk}^* - \epsilon^*) L_{kg}} \right]^{-1} d\epsilon^* \\ \epsilon_{pgk}^* &= \epsilon_2^* \left[1 + \frac{\lambda(\epsilon_1^* - \epsilon_2^*)}{\epsilon_2^* + (\epsilon_1^* - \epsilon_2^*)(L_{ik} - \lambda L_k)} \right] \\ L_{ik} &= \frac{R_{ix}R_{iy}R_{iz}}{2} \int_0^\infty \frac{ds}{(R_{ik}^2 + s)R_{is}}, \quad k = x, y, z \\ R_{is} &= \sqrt{(R_{ix}^2 + s)(R_{iy}^2 + s)(R_{iz}^2 + s)} \\ \lambda &= \frac{R_{ix}R_{iy}R_{iz}}{R_xR_yR_z} \end{aligned} \quad (7)$$

where ϕ is the total volume fraction of core and shell (which we refer to as the entire inclusion “ p ” here), r_g is the volume fraction of n different inclusion groups with different shapes relative to total inclusion volume ($\sum_{g=1}^n r_g = 1$)

(Figure 2d). ϵ_{pgk}^* is the combined complex dielectric permittivity of the g th shell-covered inclusion group. L_{kg} are depolarization factors in x, y and z directions of the g th group (reflecting the effect of inclusion shape on interac-

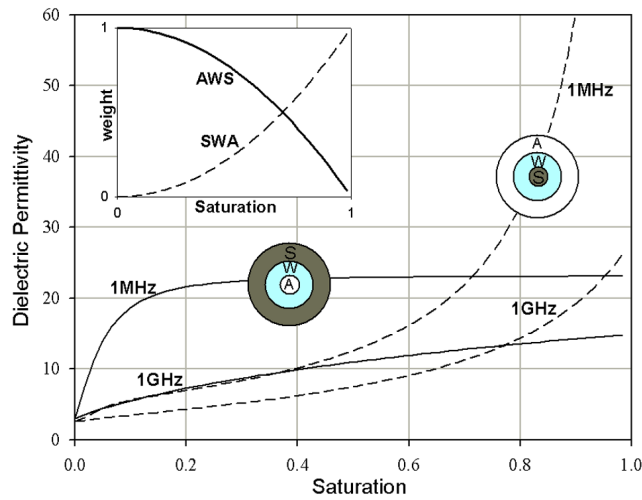


Figure 3. Dielectric permittivity versus saturation degree of hypothetical mixture composed of AWS and SWA phase configurations at frequency of 1 MHz and 1 GHz (porosity = 40% and aqueous solution electrical conductivity 0.1 S/m). Solid lines for simulated permittivity by AWS phase configuration at 1 MHz and 1 GHz. Dashed lines are simulated permittivity by SWA phase configuration at 1 MHz and 1 GHz.

tions with the electrical field). For spherical particles, L_x , L_y and L_z are 1/3 [Sen, 1984; Chelidze and Gueguen, 1999; Asami, 2002].

[13] In subsequent sections we use the simplified form of the model represented by equation (6) to simulate influences of phase configuration (section 2.2), electrical conductivity and frequency on bulk dielectric permittivity (section 2.3). To analyze effects of inclusion shape on permittivity, we use general equation (7) (section 2.4).

2.2. Effects of Fluid Phase Configuration

[14] Partial saturation is associated with complex configuration of liquid and gaseous phases in media pore space which play an important role in measured dielectric permittivity of the mixture [Sen, 1981; Chelidze and Gueguen, 1999; Cosenza et al., 2003]. Phase configuration and geometry define the types of interactions and shape of an electrical field across phase discontinuities. In the context of the proposed MWBH-DEMA model of coated inclusions, phase configuration is equivalent to defining a particular integration path for incremental phase addition to the mixture [Cosenza et al., 2003]. At a given water content and porosity whether we consider an air-water-solid (AWS) configuration (air entrapped in a water shell inclusions (bubbles) embedded in a background of solid matrix), or solid-water-air (SWA) where solid grains coated with water are placed in a background of air result in considerably different bulk permittivity, even with same types of constituents and volume fractions [Friedman, 1998; Jones and Friedman, 2000; Miyamoto et al., 2005]. As Endres and Redman [1996] and Endres [2000] have shown, the dielectric properties vary significantly with changes in fluid distribution at the pore scale. The large variations with decreasing frequency

suggest that the location of the conducting wetting phase is particularly important [Knight and Nur, 1987].

[15] For coated spheres, Friedman [1998] considered all possible combinations of the three phases and reduced these into a smaller subset on the basis of physical arguments (e.g., affinity of water to attach onto hydrophilic solid surfaces [Sen, 1981]) and symmetry considerations. Friedman [1998] eliminated candidate configurations on the basis of their deviation from Topp's model [Topp et al., 1980] model. In this study we adhered to hydrophilic interactions between solid and water and considered configurations with SW or WS combinations. Additionally, considering the broad spectrum of porous media and measurement frequencies examined here, the two end-member configuration chosen were: AWS for dielectric behavior at the low saturation range, and SWA close to saturation. We observed that the concave and steep rise in effective permittivity at the low saturation in the data of Knight and Nur [1987, Figure 10] and others can be represented only by relationships corresponding to AWS configuration, whereas SWA configuration captures the convex and steep increase in measured permittivity at high saturations (e.g., Figure 3). It is difficult to ascribe physical significance to these particular configurations other than the need for modeling the behavior with the least number of different configurations. These configurations can also be viewed as preferred sequences of phase mixing rather than reflection of physical configurations. Finally, we emphasize that experimental data provide the primary guidance in the choice of potential configurations for modeling.

[16] Additionally, the volume fractions of each of the two configurations at different saturations are unknown. Roth et al. [1990], Friedman [1998], Cosenza et al. [2003] and others have proposed the use of degree of saturation (S) as a potential physically based weight function that naturally signifies transition from one phase configuration to the other. Friedman [1998] compared models of different phase configurations with Topp's equation and proposed a linear combination of two end-member phase configurations weighted by saturation degree S . In our tests of this approach, we found that the weight function (or volume fraction) providing best fit for all data set was $\alpha = S^2$:

$$\begin{aligned} \epsilon_{eff} &= (1 - \alpha) * \epsilon_{AWS} + \alpha * \epsilon_{SWA} \\ \alpha &= S^2 \end{aligned} \quad (8)$$

[17] We evaluated more rigorous methods for combining contributions of different phase configurations to effective permittivity by considering an additional DEMA step, and simple EMA and Maxwell-Garnett mixing models discussed by Friedman [1998] and others.

[18] The DEMA step for combination of AWS and SWA two phase configurations is given by

$$\begin{aligned} \left(\frac{\epsilon_{eff} - \epsilon_{AWS}}{\epsilon_{SWA} - \epsilon_{AWS}} \right) \left(\frac{\epsilon_{AWS}}{\epsilon_{eff}} \right)^{1/3} &= \alpha \\ \alpha &= S^2 \end{aligned} \quad (9)$$

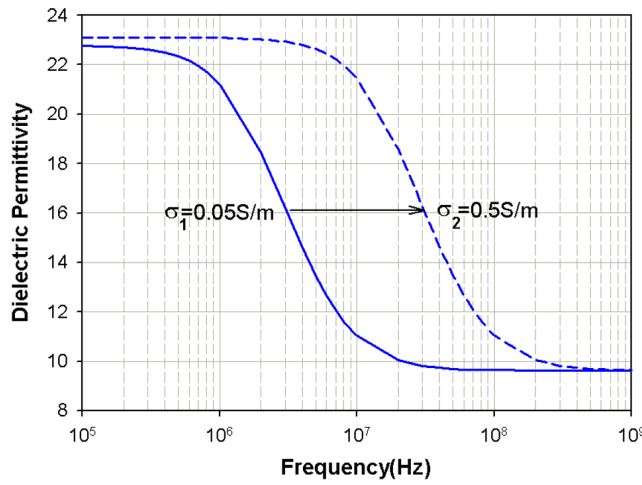


Figure 4. Model calculations of dielectric permittivity versus frequency for aqueous solutions with electrical conductivity of 0.05 and 0.5 S/m (40% porosity and 15% water content). Calculations are based on AWS phase configurations.

[19] The EMA formulation of two phases (configurations) according to *Sen* [1981] is

$$\epsilon_{eff} = \left\{ \frac{[(1-\alpha)(\epsilon_{SWA} - 2\epsilon_{AWS}) + \alpha(\epsilon_{AWS} - 2\epsilon_{SWA})]^2}{16} + \frac{\epsilon_{AWS}\epsilon_{SWA}}{2} \right\}^{1/2} - \frac{[(1-\alpha)(\epsilon_{SWA} - 2\epsilon_{AWS}) + \alpha(\epsilon_{AWS} - 2\epsilon_{SWA})]}{4}$$

$$\alpha = S^2 \quad (10)$$

[20] The *Maxwell-Garnett* [1904] model for connecting AWS and SWA phase configurations is [Friedman, 1998]

$$\epsilon_{eff} = \epsilon_{SWA} + 3(1-\alpha)\epsilon_{SWA} \left(\frac{\epsilon_{AWS} - \epsilon_{SWA}}{\epsilon_{AWS} + 2\epsilon_{SWA} - (1-\alpha)(\epsilon_{AWS} - \epsilon_{SWA})} \right)$$

$$\alpha = S^2 \quad (11)$$

[21] We evaluated these four different mixing models for a wide range of experimental data. The use of $\alpha = S^2$ was found to be superior to $\alpha = S$ for all models tested, in particular for the simple linear weight function (equation (8)).

2.3. Effects of Electrical Conductivity and Measurement Frequency

[22] Typically, soil minerals are considered as nonconductive (or dielectric insulators) whose electrical conductivity is less than 10^{-10} S/m. In contrast, soil solution often spans a wide range of electrical conductivities dominating the mixture electrical conductivity and giving rise to enhanced M-W effect with a significant frequency dependency of measured dielectric permittivity. We distinguish between soil solution electrical conductivity which, according to the model, directly influences measured bulk permittivity of the mixture, and the bulk electrical conductivity of the soil mixture.

[23] Illustration of the relationships between electrical conductivity and frequency-dependent permittivity with

the MWBH-DEMA model (equations (6) and (7)) is complicated by the 1/3 power and integrations, these are more clearly shown using the Wagner model for dilute inclusion concentration (Figure 2a) considering hypothetical two-phase porous medium [Sherman, 1968]. Assuming a mixture with solid inclusions with 0 electrical conductivity and ϵ_s permittivity, embedded in a background liquid with σ_l electrical conductivity and ϵ_l permittivity (Figure 2a), the frequency-dependent permittivity of this mixture is explicitly given by

$$\epsilon^* = \epsilon_l \frac{2(1-\phi_s)\epsilon_l + (1+2\phi_s)\epsilon_s}{(2+\phi_s)\epsilon_l + (1-\phi_s)\epsilon_s} + \frac{\Delta\epsilon}{1+j\omega\tau} + \frac{\sigma^*}{j\omega\epsilon_0}$$

$$\Delta\epsilon = \frac{9\phi_s(1-\phi_s)\epsilon_s^2}{(2+\phi_s)^2[2\epsilon_l + \epsilon_s + \phi_s(\epsilon_l - \epsilon_s)]} \quad (12)$$

$$\tau = \frac{1}{2\pi f_r}, \quad f_r = \frac{(2+\phi_s)}{2\pi\epsilon_0[2\epsilon_l + \epsilon_s + \phi_s(\epsilon_l - \epsilon_s)]} \sigma_l$$

$$\sigma^* = \frac{2(1-\phi_s)}{(2+\phi_s)} \sigma_l$$

where ϵ_0 is the dielectric permittivity of free space (8.8541×10^{-12} farad/m), ϕ_s is the volumetric fraction of the solid inclusions. ω is the angular frequency and has the following relationship with frequency: $\omega = 2\pi f$, $j = \sqrt{-1}$. τ is the relaxation time, and f_r is the M-W relaxation frequency. Equation (12) highlights the dependency of bulk dielectric permittivity on relaxation frequency, which, in turn, is dependent on solution's electrical conductivity σ_l .

[24] Next, the MWBH three-phase model (equation (6)) is used to simulate the influence of solution's electrical conductivity on the bulk permittivity. It is assumed that the soil mixture has 0.40 porosity and 0.15 volumetric water content. The permittivity of soil solids is 5, and 1 for air (both with 0 electrical conductivities), and soil solution permittivity is 78.4 and the bulk permittivity was simulated for two soil water electrical conductivities of 0.05 S/m and 0.5 S/m (Figure 4). The results illustrate the migration of the M-W relaxation to higher frequencies with increasing electrical conductivity (e.g., from 3 to 30 MHz). Similar changes in electrical conductivity of aqueous solution in a porous medium could affect the measured values of bulk permittivity for similar water contents. In view of the ubiquity of ionic charges in soils and presence of numerous interfaces, the extended range of M-W polarization may have significant impact on dielectric permittivity measurements even for sensors operating at relatively high frequencies (>100 MHz) such as TDR.

2.4. Effects of Inclusion Shape

[25] Several studies have focused on the impact of solid/pore shape on measured and simulated dielectric permittivity [Feng and Sen, 1985; Jones and Friedman, 2000]. To evaluate the role of inclusion shape in our modeling scheme, we compared the effective permittivity for spherical inclusions (single depolarization factor = 1/3) with the permittivity for ellipsoidal inclusions (depolarization factors of: 0.1, 0.1, 0.8) for the same volume fraction of randomly oriented inclusions. The results show large differences between spherical and ellipsoidal inclusions on mixture permittivity at low frequencies (Figure 5), however, shape effects become less important at higher frequencies (50 MHz curves in Figure 5).

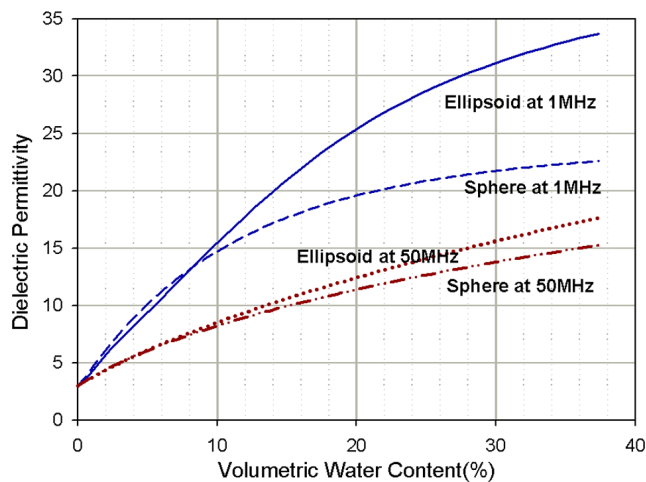


Figure 5. Dielectric permittivity versus water content at frequency of 1 and 50 MHz comparing spherical and ellipsoidal inclusions (assuming 40% porosity, AWS configuration, and 0.1 S/m aqueous phase electrical conductivity).

[26] In subsequent sections, we consider inclusions of spherical shape for sandy or silt loam soil samples (sections 3.1 and 3.3). For the clay samples, we compare simulations using spherical and ellipsoidal shapes, whose depolarization factors varied to provide best match with measurements (in the absence of clay geometrical information).

3. Model Application to Experimental Data

[27] Theoretical simulation results illustrated the respective roles of phase configuration, electrical conductivity, and the M-W effect on the frequency-dependent dielectric behavior of wet porous media, especially at the low-frequency range (kilohertz–megahertz). To evaluate the importance of these processes in practice, we apply the modeling framework to experimental data reported in the literature spanning a wide range of water-bearing porous media, saturation degrees, mineralogy, and porosities. Recognizing limitations of these types of tests, our primary interest lies in identification of unifying themes that could implement and improve our capabilities to interpret dielectric behavior of real porous media measured by various sensors and measurement frequencies.

3.1. Dielectric Permittivity of Low-Clay Content Porous Media

[28] *Campbell* [1990] has used a network analyzer and a coaxial sample holder to measure the dielectric permittivity of partially saturated Wilder silt soil (porosity = 0.41) reported at frequencies of 1 MHz, 5 MHz and 50 MHz. A constant electrical conductivity value of 0.05 S/m for the soil solution was assumed for all calculations. The soil bulk permittivity was modeled using two phase configurations, AWS and SWA weighed by saturation squared as weight function, and combined by four different schemes summarized by equations (8)–(11) (section 2.2).

[29] Model calculations using different approaches to combining effective permittivities of end-member phase configurations were evaluated with measured data sets. A sample comparison for *Campbell's* [1990] Wilder silt soil

measurements is depicted in Figures 6a and 6b illustrating the generally superior performance of the Maxwell-Garnet (M-G) scheme (equation (11)) with a single weight $\alpha = S^2$ providing best fit to most data evaluated. Linear combination (equation (8)) provided better match than DEMA and EMA schemes, consequently, we retained the M-G and linear combination of phase configurations for subsequent evaluation of modeling results.

[30] Results depicted in Figure 6 show a sharp increase in dielectric permittivity with the introduction of small amounts of water into the dry soil (<5% water content) for both 1 MHz and 5 MHz frequencies. A rapid increase of dielectric permittivity at low water contents is characteristic of AWS phase configuration (Figure 3).

[31] *Knight and Nur* [1987] measured the relative dielectric permittivity of partially saturated Berea sandstone at frequencies of 60 KHz, 105 KHz, 472 KHz and 1.2 MHz. The porosity of Berea sandstone is about 0.2, and the clay content is low. The electrical conductivity of the solution

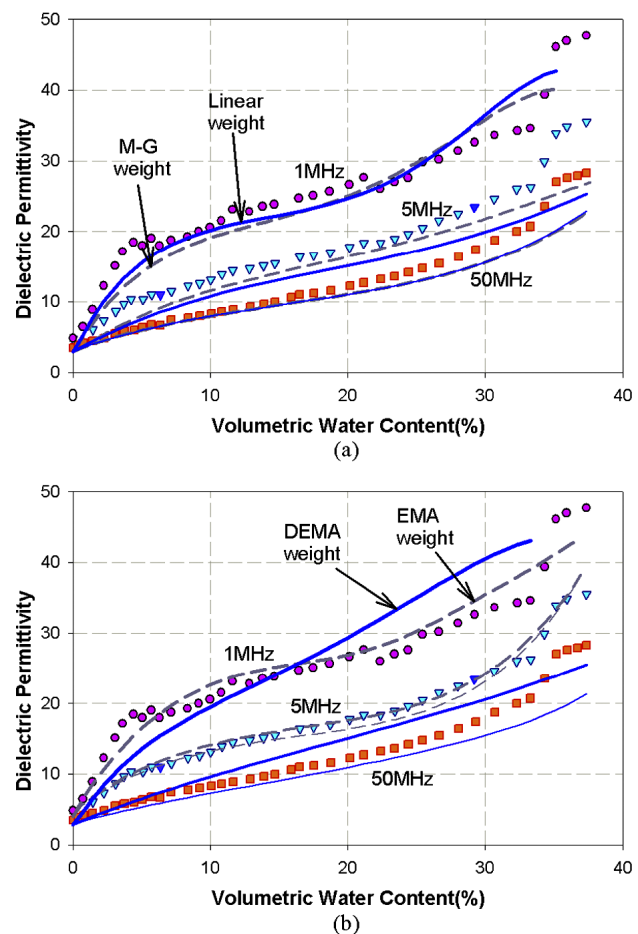


Figure 6. Measurements (symbols) and model simulations (lines) of dielectric permittivity versus volumetric water content for Wilder silt soil [*Campbell*, 1990]. (a) Simulated permittivity by linear weight of AWS and SWA phase configurations (equation (8), solid lines) and simulated permittivity by Maxwell-Garnett (M-G) weight function (equation (11), dashed lines). (b) Simulated permittivity by DEMA scheme (equation (9), solid lines) and simulated permittivity by EMA scheme (equation (10), dashed lines).

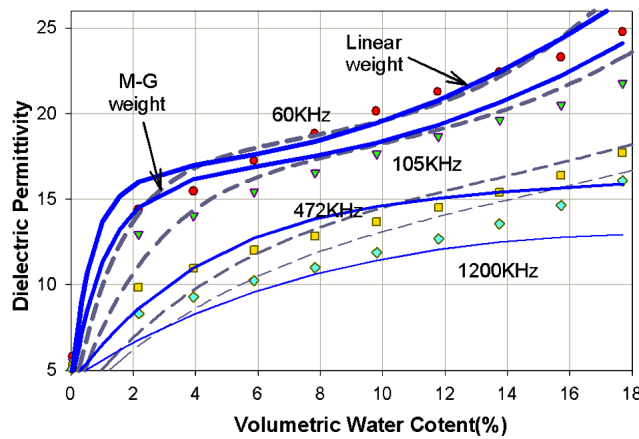


Figure 7. Measured (symbols) and simulated (lines) dielectric permittivity versus volumetric water content at four frequencies for the Berea 100 sandstone data reported by *Knight and Nur* [1987]. Solid lines are the simulated permittivity by linear weight function (equation (8)); dashed lines are the simulated permittivity by Maxwell-Garnett (M-G) weight function (equation (11)).

was not reported and a value of 0.008 S/m was assumed. The simulated results with AWS and SWA configurations combined with the saturation degree weight function (equations (8) and (11)) match measured values quite well as depicted in Figure 7.

[32] Summarizing, dielectric measurements in porous media with low clay content show a sharp rise of dielectric permittivity at low water content at low measurement frequency. The magnitude of the rise decreases with increasing measurement frequency. The best match with our modeling approach is obtained with AWS as the dominant phase configuration at low saturation. This could be explained by wetting patterns of low-porosity sandy soils and sandstones occurring by association with solid surfaces and crevices first without complete expulsion of air (incomplete pore filling) followed by gradual filling of large pores at higher saturations.

3.2. Permittivity of Porous Media With High Clay Content

[33] *Saarenketo* [1998] measured the dielectric behavior of Houston Black clay and Beaumont clay. Because of the high clay content in these soils, we considered the representation of ellipsoidal inclusions which require geometrical parameters (equation (7)) that are typically unavailable. To assess the need for accommodating shape effects, we assumed the inclusion is oblate spheroid with $R_x = R_y = 100R_z$. The porosity of Houston Black clay is assumed 0.6, and solution electrical conductivity is 1.5 S/m [*Cosenza et al.*, 2003]. Simulation results were in very good agreement with measured values (Figure 8, lines). Similar results were obtained for Beaumont clay (porosity = 0.6 and solution electrical conductivity of 0.8 S/m [*Cosenza et al.*, 2003]) using ellipsoidal inclusions (Figure 9). All simulations consider AWS and SWA phase configurations weighted with by saturation degree using equations (8) and (11).

[34] In summary, the proposed model and phase configuration weight function resulted in good agreement with dielectric measurements of soils with appreciable clay

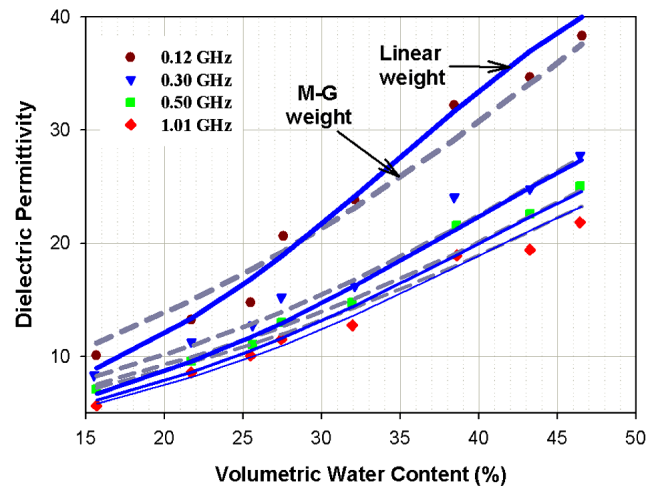


Figure 8. Measured (symbols) and simulated (lines) dielectric permittivity versus volumetric water content at four frequencies based on Houston Black clay data reported by *Saarenketo* [1998]. Solid lines represent simulated permittivity by linear weight function; dashed lines represent simulated permittivity by Maxwell-Garnett (M-G) weight function.

content at different measurement frequencies. The nonlinear response of dielectric permittivity behavior at higher water contents is attributed to dominance of SWA phase configuration. The associated mechanism of pore filling behind this configuration is less obvious, it may be related to water affinity to large surface area and preference of SW association coupled with swelling of saturated aggregates and gradual displacement of air phase.

3.3. Role of Electrical Conductivity in Saturated Porous Media

[35] Low-frequency M-W polarization affecting dielectric behavior results from the presence of nonuniformly distrib-

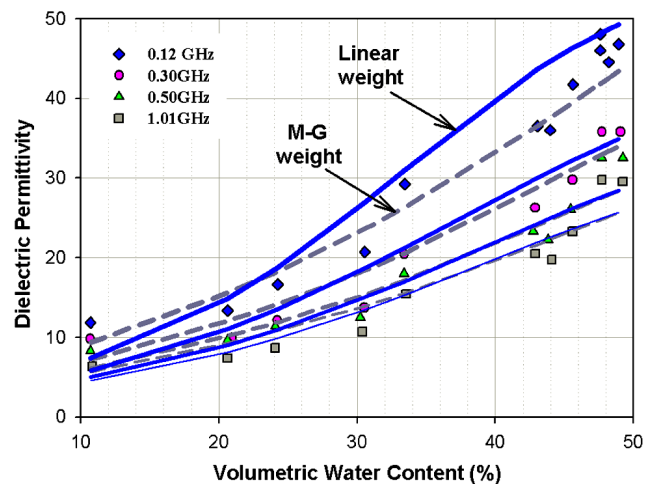


Figure 9. Measured (symbols) and simulated (lines) dielectric permittivity versus volumetric water content at four frequencies for Beaumont clay data reported by *Saarenketo* [1998]. Solid lines represent permittivity calculated by linear weight function, and dashed lines represent permittivity by Maxwell-Garnett (M-G) weight function.

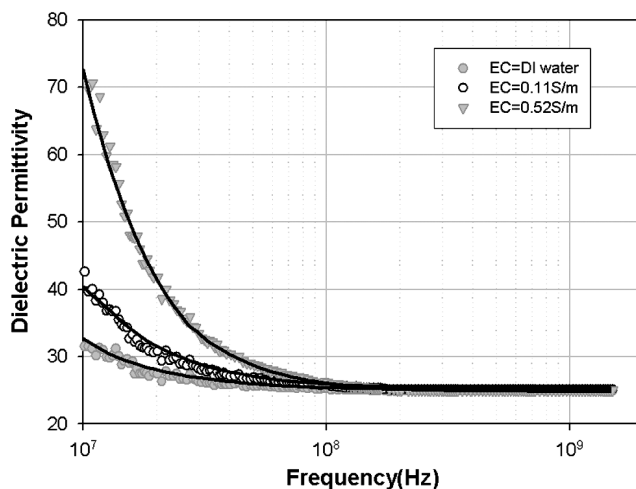


Figure 10. Measured (symbols) and simulated (lines) dielectric spectra for Woodbridge silt loam saturated with aqueous solutions with three different electrical conductivity values of DI water and 0.11 and 0.52 S/m (porosity $\sim 44\%$ and water content $\sim 41\%$).

uted charges and formation of numerous interfaces [Knight and Nur, 1987].

[36] To isolate and verify the role of electrical conductivity on permittivity measurements, we conducted experiments using sieved Woodbridge silt loam (particles < 2 mm sieve with a mean particle size of 0.1 mm and porosity of 0.44). The samples were saturated (attained volumetric water contents of ~ 0.41) using deionized (DI) water, and two NaCl solutions with electrical conductivities of 0.11 S/m and 0.52 S/m. The complex permittivity was measured by a network analyzer (Agilent 8753D Network Analyzer equipped with an 85070E high-temperature dielectric probe). The high-frequency average values of the real part of saturated sample permittivities for DI water, 0.11 S/m and 0.52 S/m solutions were 24.1, 23.8 and 23.8, respectively (averages for frequencies above 1 GHz; see Figure 10). Simulations of the frequency-dependent permittivities (focusing on the real part) of the saturated soil samples using the MWBH-Asami model resulted in very good agreement with the trends exhibited by the measurements (Figure 10, lines). The simulated permittivity value at frequency above 1 GHz was 25.9, similar for all three samples.

[37] The results depicted in Figure 10 clearly show the effect of electrical conductivity on the M-W polarization in saturated porous media. These tests confirm the expected increase in bulk dielectric permittivity that would be measured at a fixed frequency (e.g., 10 MHz) with increasing electrical conductivity. We also observe the migration of dielectric relaxation associated with the M-W effect to higher frequencies with increasing electrical conductivity of the aqueous solution. Finally, the role of measurement frequency is evident in all measurements, showing a pronounced decrease in measured (and modeled) dielectric permittivity with increasing frequency due to gradual relaxation of interfacial phenomena.

4. Summary and Conclusions

[38] The primary objective of this study was to develop modeling tools for simulating and verifying frequency-

dependent dielectric behavior of partially saturated porous media with emphasis on the low-frequency range (< 500 MHz) where M-W effect is significant. We have used the MWBH-Asami formulation to predict complex permittivity of a three-phase porous medium composed of coated spherical and ellipsoidal inclusions. The model explicitly accounts for the roles of phase configuration, inclusion shape, electrical conductivity (EC), and frequency where the Maxwell-Wagner polarization affects mixture dielectric permittivity.

[39] Modeling results and interpretation of data reported in the literature show the dominant roles of phase configuration and interfacial processes affecting the dielectric permittivity spectrum of wet soils especially at frequencies below 100 MHz. This dependency highlights limitations of standard measurement interpretation on the basis of Topp's equation. Experimental results and physical considerations enable reduction of the number of phase configurations to AWS and SWA configurations only. Moreover, the use of a single saturation-based weight function with simple M-G or linear combination of phase configurations provided a good match with measured permittivity across many soils types, saturation degrees, and frequencies. Results show that particle or inclusion shapes become important for clayey soils especially at lower frequencies. Typically, the bulk dielectric permittivity at low frequency increases with increasing electrical conductivity due to the M-W effect. This effect may extend to higher frequencies (> 100 MHz range) well within the measurement range of TDR and similar sensors. The model developed in this study establishes direct links between modeled complex dielectric spectra and measured TDR waveforms thereby enhancing measurement interpretation capabilities for TDR and new sensors under different soils and ambient conditions. The unifying framework admits incorporation of additional effects such as bound water and temperature that will be investigated in future studies.

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